

Problem Set 8: Cell Cycle and Signaling

Due: Sunday, December 6 at 11:59 PM

Value: 25 points

Format: Five-question problem set

Purpose

Connect receptor activation, signal amplification, feedback, checkpoints, and cancer-related mutations.

Your task

- Q1 (5): Draw a receptor-to-transcription pathway containing a receptor, relay, kinase, transcription factor, and negative regulator.
- Q2 (5): Explain how a small amount of ligand can produce a large response and name two steps where amplification can occur.
- Q3 (5): Predict the phenotype of a receptor that is active without ligand and a kinase that cannot be phosphorylated.
- Q4 (5): Compare the G1/S and spindle checkpoints by what is monitored and what failure risks.
- Q5 (5): Use the cell-count data to decide whether Drug X causes arrest, death, or both, and name one additional measurement needed.

Provided evidence or data

Condition	G1	S	G2/M	Dead cells
Control	48%	31%	21%	4%
Drug X	18%	22%	60%	15%

Submit

- One PDF with a labeled signaling diagram and evidence-based answers.

Grading criteria

Criterion	Pts	Full-credit evidence
Question 1	5	Correct setup, units, and biological interpretation.
Question 2	5	Correct reasoning shown; not just a final answer.
Question 3	5	Mechanism-based explanation uses appropriate evidence.
Question 4	5	Prediction follows from the stated model and conditions.
Question 5	5	Data are represented or evaluated accurately.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.